Pharmacology I

เภสัชวิทยา 1

Drug interactions

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ภาควิชาเภสัชวิทยา คณะเภสัชศาสตร์ มหาวิทยาลัยมหิดล

วัตถุประสงค์รายวิชา

เมื่อจบการเรียน นักศึกษาสามารถ

1. เข้าใจปัจจัยที่ส่งผลต่อการเกิดอันตรกิริยาของยา
2. สามารถแยกแยะประเภทของการเกิดอันตรกิริยาของยาได้
3. สามารถอธิบายกลไกและผลของเภสัชจลนศาสตร์ต่อการเกิดอันตรกิริยาของยา
4. สามารถอธิบายกลไกและผลของเภสัชพลศาสตร์ต่อการเกิดอันตรกิริยาของยาได้

Introduction

- **A drug interaction (DI) (อันตรกิริยา) can be defined as**
  - **the modifications of effects of one drug by the administration of other substances**

    - drugs
    - nutrients (food)
    - herbs
    - smoking or alcohol

- **The drug interaction which results into an undesired effect (adverse drug interaction)**

Factor affecting Drug interaction

- **Polypharmacy**
  - The rate of DI increases with the number of new drugs.

- **Genetic factors**
  - Drug metabolize enzyme are synthesis by gene: **poor metabolizer, rapid metabolizer**
  - Genotypic variation could decrease/increase the activity of these enzyme.
  - The variation of cytochrome P450 would lead to predisposition for adverse effects.

- **Liver or Kidney disease**
  - Liver or Kidney diseases affect
    - the metabolism
    - elimination of drugs
- **Old age**
  - The normal physiological functions decrease with rising age (Old age) (liver metabolism, kidney function, neuronal function).
  - A decreased sensory increase the chance of errors in the administration of drugs.

Factor affecting Drug interaction: polypharmacy

Table 5: Predictors of D/Is (multivariate logistic regression analysis).

Variables	Groups	Patients with D/Is	Patients without D/Is	Wald	Odds ratio (95% CI)	*p* value
Age	<70	75	22	4.04	1.05 (1.00–1.097)	0.04
	>70	99	13			
Gender	Male	89	15	0.18	0.86(0.42–1.75)	0.67
	Female	85	20			
Number of comorbidities	1	22	7	0.31	0.87 (0.54–1.42)	0.58
	2	52	15			
	3	75	10			
	4	21	2			
	>4	4	1			
Type of comorbidity	Diabetes	136	27	0.20	0.81 (0.31–2.11)	0.66
	Hypertension	130	23	1.71	1.92 (0.72–5.09)	0.19
	CAD	21	1	0.60	1.56 (0.51–4.82)	0.44
	Dyslipidemia	14	1	0.05	1.17 (0.31–4.46)	0.82
	CVA	51	7	0.38	0.76 (0.32–1.81)	0.54
Number of medications	<6	48	23	16.47	10.37 (3.35–32.11)	<0.001
	>6	126	12			

Varsha Shetty. Journal of Aging research. 2018; ID5728957: DOI-10.1155/2018/5728957

Factor affecting Drug interaction

- Other underlying diseases
    - Cardiovascular disease, diabetes, kidney disease, liver disease
- Lifestyle
    - Diet
    - Exercise
- The duration of combined therapy

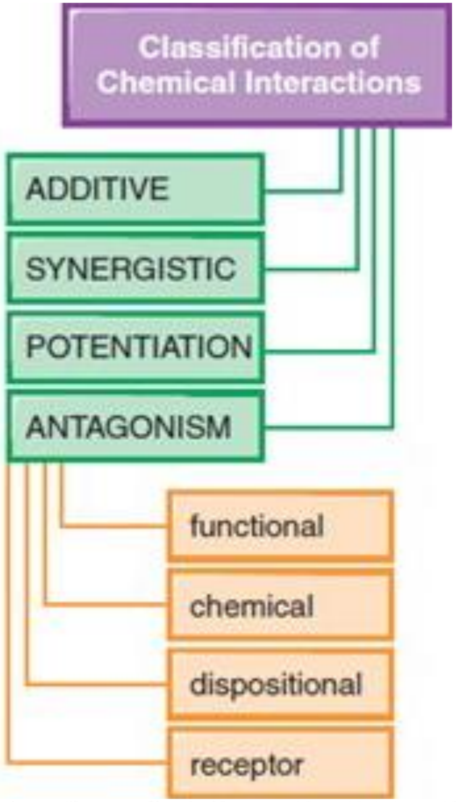
- Therapeutic index of drugs
    - Narrow therapeutic index
        - The difference between the effective dose and the toxic dose is small.

Consequences of Drug interaction

- Increased beneficial effects
  - Combining two or more drugs acting in different mechanisms may produce better results than with either drug alone.

- Decreased beneficial effects
  - Drugs reducing absorption or increasing metabolism or elimination of other drugs, cause reduced therapeutic efficacy.

- Increased adverse effects
  - Drugs increasing absorption or reducing metabolism or elimination of other drugs, produce more side effects by increasing their plasma levels.
  - When two drugs that produce similar side effects are combined, the frequency and severity of the side effects are increased.



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Type of drug interaction

- Drug-Drug interactions
- Herb-Drug interactions
- Nutrient (Food)-Drug interactions
- Smoking-Drug interactions
- Alcohol-Drug interactions
- Disease-Drug interactions

Drug-Drug interaction

- **The modification of the effect of a drug when administered with another drug**
  - **Interactions can occur with prescription drugs, over-the-counter drugs, etc.**
    - **Sedative (Prescription drug) + Antihistamine (OTC antiallergic)**
      - **Daytime drowsiness**
      - **Affects driving or operating machinery**

Herb-drug interaction

- **May increase or decrease the pharmacological or toxicological effects of either component.**

  - **St Johns wort** (Hypericum perforatum) + **Digoxin, Theophylline or Cyclosporin**
    - **St Johns wort is a potent inducer of cytochrome P450 isoenzymes**
      - **Increased metabolism of drugs**
      - **Decreased bioavailability of digoxin, theophylline and cyclosporin**

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Nutrient-drug interaction

- Drug-food interactions occur when a drug interacts with food or drinks.

  - Dairy products, such as milk, yogurt and cheese
    - Provides Ca2 ions
    - Chelate with Tetracyclines
    - Decreased absorption of antibiotics into the bloodstream

  - Vegetables (Broccoli and spinach)
    - containing vitamin K
    - Decrease the effectiveness of Anticoagulant Coumadin (warfarin)

Smoking-drug interaction

- **Tobacco smoke interacts with medications through pharmacokinetic (PK) and pharmacodynamic (PD).**
    - **PK interactions affect the absorption, distribution, metabolism, or elimination of other drugs,**
        - **Most PK interactions with smoking are the result of induction of hepatic cytochrome P450 enzymes (primarily CYP1A2).**
    - **PD interactions alter the expected response or actions of other drugs.**
        - **Tobacco smoke**
        - **Induction of CYP1A2**
        - **Increased metabolism of Clopidogrel (ยาต้านเกล็ดเลือด)**
            - **Induced formation of active metabolites of clopidogrel**
            - **significant rise in platelet inhibition**
            - **Improved clinical outcomes and may also increase risk of bleeding**

Alcohol-drug interaction

- Mixing alcohol with some drugs is dangerous.

  - Alcohol (beer, wine, or liquor) interacts with
    - most antidepressants and with other drugs that affect the brain.
      - the combination can cause fatigue, dizziness, and slow reactions.
    - OTC NSAIDs (Aspirin, Ibuprofen, and Acetaminophen)
      - increased risk of stomach bleeding or liver damage

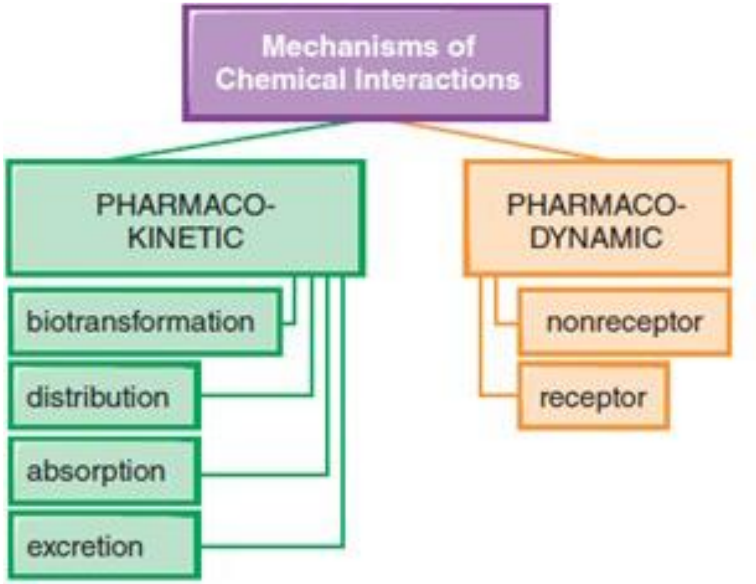
Disease-drug interaction

- may occur when a medication interacts with an existing health condition.

    - Propronolol VS Asthma or COPD
        - Propronolol cause bronchoconstriction
            - non-specific binding of drug to receptors
            - Aggravation of symptoms of Asthma and COPD

Mechanism of drug interaction

- Pharmaceutical Interactions
- Pharmacokinetic interactions
    - Absorption
    - Distribution
    - Metabolism
    - Excretion
- Pharmacodynamics interactions
    - Receptor mediated
    - Non receptor mediated



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Mechanism of DI: Pharmaceutical DI

- Pharmaceutical Interactions are
  - chemical reactions occur when two or more drugs are mixed outside the human body for the purpose of joint administration.

  - Pharmaceutical interactions are also known as pharmacological incompatibilities.
    - Penicillin VS. Aminoglycosides
      - Mixing in a same bottle
      - Formation of an insoluble precipitate
      - Loss of therapeutic efficacy of both the drugs

Mechanism of DI: PK interaction

- **Pharmacokinetic interactions are those in which one drug alters the concentration of another drug (increase or decrease) in the system.**
  - **Pharmacokinetic interactions are more complicated and difficult to predict because the interacting drugs often have unrelated actions.**
    - **Affect drugs bioavailability, volume of distribution, peak level, biotransformation, clearance and half-life**
    - **Changes in drug plasma concentrations**
    - **Increased risk of side effects or diminished therapeutic efficacy of one or more drugs**

Mechanism of DI: PD-interaction

- Pharmacodynamic interactions are those in which the effect of one drug is changed by

  - the presence of another drug acting at
    - the same biochemical or molecular site,
    - on the same target organ,
    - on a different target.

These PD-drug interaction can be classified to be:

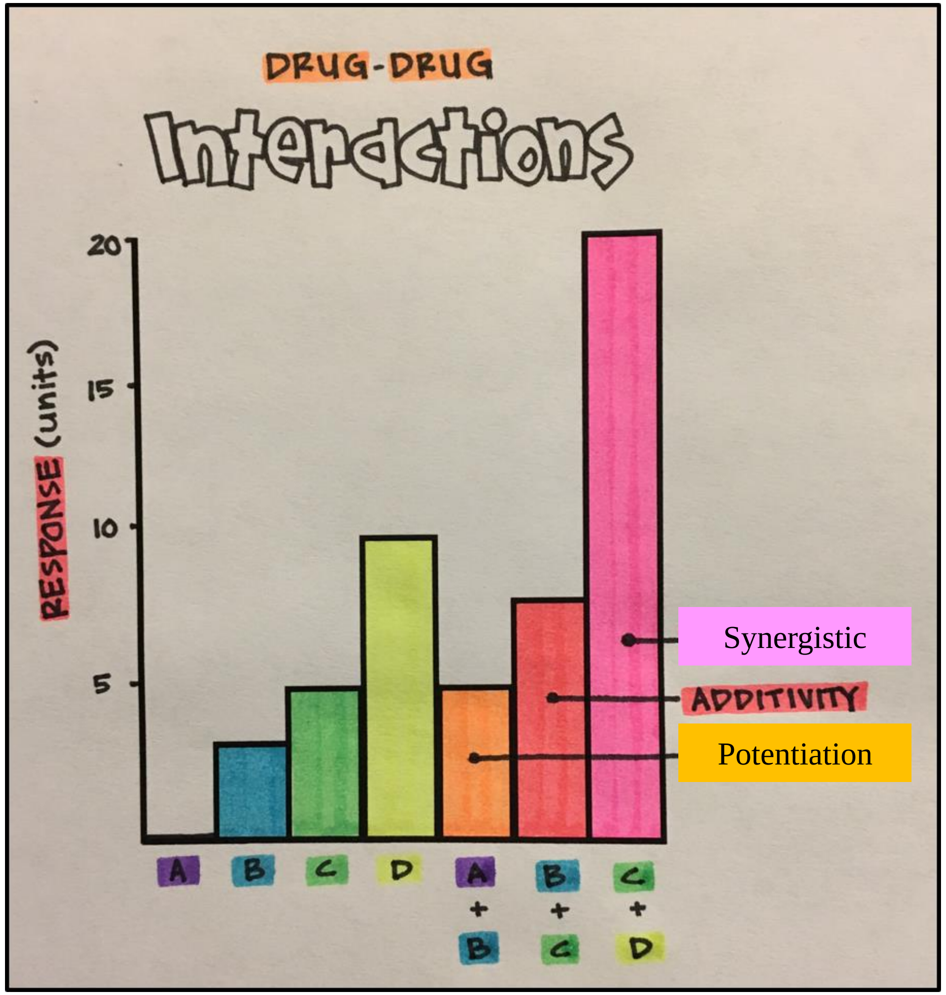
- Receptor mediated

- Non receptor mediated

Classification of drug interaction

- **Additive**
  - **1+1 =2**
- **Synergistic**
  - **1+1 >>>2**
- **Potentiation**
  - **a drug with no effect is combined with another drug to cause an effect**

- **Antagonism (1+1 <2)**
  - **one drug weakens or cancels out the effect of another drug**
    - Functional
    - Chemical
    - Dispositional
    - Receptor



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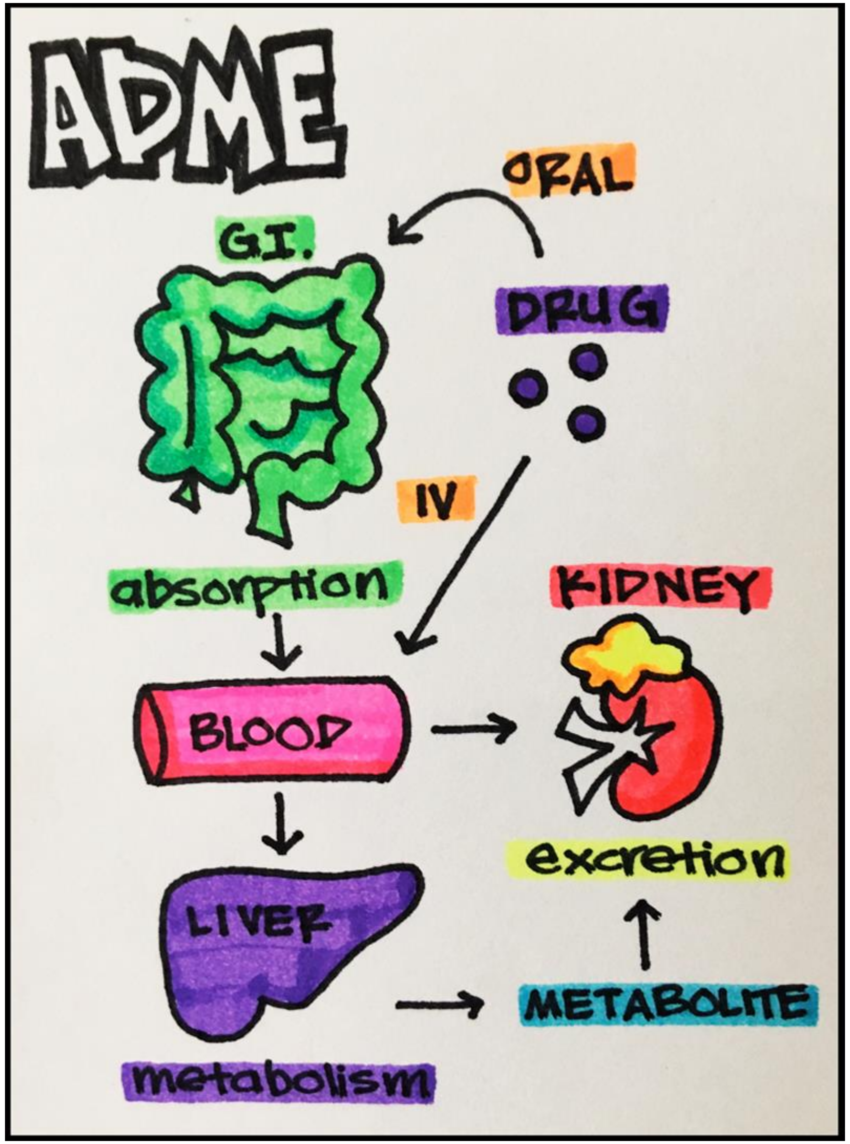
Pharmacokinetic drug interaction

- Pharmacokinetic interactions involve in the alteration of drugs
  - Absorption
  - Distribution
  - Metabolism
  - Excretion



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PK drug interaction: absorption

- Drug may cause either an increase or a decrease in the absorption of another drug from the intestinal lumen.

- Absorption of some drugs are altered by the presence of other drugs due to
  - Changes in gastrointestinal pH
  - Adsorption, chelation and other complexing mechanisms
  - Changes in gastrointestinal motility
  - Induction or inhibition of drug transporter proteins
  - Malabsorption caused by drugs

Absorption: change in GI pH

- At low pH (acidic pH), the absorption of acidic drugs like salicylic acid, etc is high
  - but their absorption is reduced due to a rise in pH (basic pH).
  - Proton pump inhibitors (PPIs) such as Omeprazole, Esomeprazole, Pantoprazole, Rabeprazole
  - H2 receptor blockers like Ranitidine

    Tend to reduce the absorption of acidic drugs.

    - PPIs or H2-receptor blockers
      - Reduce gastric acid secretion
      - Increased gastric pH
      - Poor dissociation of ketoconazole (poorly soluble base) due to less acidic environment
      - Decreased absorption of Ketoconazole

Absorption: change in GI pH (base)

- At high pH (basic pH), the absorption of basic drugs like triazolam is high
    - but their absorption is reduced due to a reduction in pH (acidic pH)

    - Ranitidine (H2 receptor blocker)
        - Reduces gastric acid secretion
        - Increases gastric pH
        - Increases the absorption of Triazolam

Absorption: inhibit

- Activated charcoal or Antacids
    - Bind a large number of drugs
    - Reduced absorption

Absorption: Chelation

- Chelation and other complexing mechanisms

  - Tetracyclines
    - Chelate with divalent and trivalent metallic ions, such as calcium, aluminium, bismuth and iron of antacids and dairy products
    - Formation of complexes
    - Poor Absorption
    - Reduced Antibacterial effects

Absorption: Chelation

- Bile acid sequestrants (Cholestyramine)
  - Formation of complexes with Digoxin or warfarin or levothyroxine
  - Reduced absorption

Absorption: GI motility (decrease)

- **The absorption of drugs such as Paracetamol, etc. is reduced by the presence of drugs like propantheline (Antimuscarinics), etc. which affect the gastric emptying.**
  - **Propantheline (Antimuscarinic)**
    - **Block M3 receptors of smooth muscle of GIT**
    - **Reduce contraction of smooth muscles**
    - **Decreased GI motility**
    - **Delayed gastric emptying**
    - **Reduced rate of absorption of Paracetamol (acetaminophen)**

Absorption: GI motility (decrease)

- Drugs with antimuscarinic effects (Tricyclic Antidepressants, etc.)

  - Block M3 receptors of smooth muscle of GIT
    - Reduce contraction of smooth muscles
    - Decreased GI motility
    - Delayed gastric emptying
    - Reduced absorption of Levodopa

Absorption: GI motility (increase)

- The absorption of drugs such as Paracetamol, etc. is increased by the presence of drugs like Metoclopramide, Domperidone, etc. which affect the gastric emptying.

  - Metoclopramide or Domperidone
    - Block D2 receptors
    - Increased GI motility
    - Raise the gastric emptying
    - Increased rate of absorption of Paracetamol (acetaminophen)

Absorption: Drug transporter (induce)

- Drug transporter proteins such as P-glycoprotein, determine the oral bioavailability of some drugs by ejecting drugs that have diffused across the gut lining back into the gut.

  - Rifampicin (Rifampin)
    - Induce P-glycoprotein
    - Eject digoxin into gut more vigorously
    - Reduced absorption of Digoxin
    - Fall in the plasma levels of digoxin

Absorption: Drug transporter (inhibit)

- Verapamil (calcium-channel blocker)
- Inhibit the P-glycoprotein-mediated transcellular transport of digoxin
- Inhibiting the renal and extra renal (Biliary) excretions of digoxin
- Increased digoxin levels

Absorption: malabsorption

- **Neomycin**
    - **General malabsorption syndrome**
    - **Impair the absorption of drugs like digoxin, and methotrexate**

PK drug interaction: Distribution

- Drugs distribution is affected by
  - Protein-binding interactions
  - Induction or inhibition of drug transporter proteins

Distribution: Protein binding

•The binding of drugs to the plasma proteins is reversible.

•The bound and unbound molecules are established at equilibrium.

•Only the unbound molecules remain free and pharmacologically active.

•Bound molecules are pharmacologically inactive and are temporarily protected from metabolism and excretion.

•As the free (Unbound) molecules become metabolised, some of the bound molecules become unbound and pass into solution to exert their normal pharmacological actions.

Distribution: Protein binding

- Warfarin VS. Chloral hydrate

- Chloral hydrate
  - Trichloroacetic acid (Major metabolite)
  - Displaces warfarin from binding
  - Free and active warfarin
  - Exposed to metabolism
  - Very short lived effects

Distribution: drug transporter

- Drug transporter proteins such as P-glycoprotein limit the distribution of drugs into the tissues.

- These proteins actively transport drugs out of cells when they have passively diffused in.
    - Drugs that are inhibitors of these transporters could therefore increase the uptake of drug substrates into the tissue, which could either increase adverse effects, or be beneficial.

Distribution: drug transporter (induction)

- Induction of P-glycoprotein
  - Rifampicin (Rifampin)
    - Stimulation of P-glycoprotein within the lining cells of the gut
    - Ejects Digoxin into the gut more vigorously
    - Fall in the plasma levels of Digoxin

Distribution: drug transporter (inhibition)

- Inhibition of P-glycoprotein
    - Ketoconazole
        - Inhibition of P-glycoprotein
        - Prevention of efflux of Ritonavir from CNS
        - Increase the CSF levels of ritonavir

Distribution: drug transporter(inhibition)

- Inhibition of P-glycoprotein

  - Verapamil
    - Inhibit the activity of P-glycoprotein
    - Prevents the ejection of Digoxin into the gut
    - Increased Digoxin levels

PK drug interaction: Metabolism

- The main enzymatic system responsible for drug metabolism is the cytochrome P-450 (CYP) system.
- Metabolism through the CYP system occurs mainly in the liver, but CYP isozymes are also found in the intestines and other organs.

  - There are six isozymes for most known drug interaction CYP 1A2, 2C9, 2C19, 2D6, 3A4, and 2E1.
    - There are three ways in which a drug can interact with the isozymes
      - Substrate
        - Drug is metabolized by an isozyme that is specific for an individual CYP receptor.
      - Inducer
        - Drug "revs up" the isozyme system, allowing a greater metabolism capacity.
      - Inhibitor
        - Drug(s) competes with another drug(s) for a specific isozyme-binding site, rendering the isozyme inactive.

PK drug interaction: Metabolism

- Changes in first-pass metabolism
    - Changes in blood flow through the liver
    - Inhibition or induction of first-pass metabolism
- Enzyme induction
- Enzyme inhibition
- Genetic factors in drug metabolism
- Cytochrome P450 isoenzymes and predicting drug interactions

Metabolism: change first-pass metabolism

- Changes in blood flow through the liver

- The drugs are taken to the liver after absorption in the intestine, by the portal circulation before they are distributed by the blood flow around the rest of the body.

- A number of highly lipid-soluble drugs undergo substantial biotransformation during this first-pass through the gut wall and liver.

- Verapamil
  - Increase hepatic blood flow
  - Increased rate of absorption of dofetilide
  - Increased dofetilide plasma levels
  - QTc prolongation
    - Increased risk of Torsade de pointes

Metabolism: changes first-pass metabolism

- Inhibition or induction of first-pass metabolism

- The gut wall contains metabolizing enzymes, principally cytochrome P450 isoenzymes.
- Some drugs can have a marked effect on the extent of first-pass metabolism by inhibiting or inducing the cytochrome P450 isoenzymes in the gut wall or in the liver.

- Grapefruit juice
    - Inhibits the cytochrome P450 isoenzyme CYP3A4 (mainly in the gut)
    - Reduction of the metabolism of oral calcium-channel blockers
    - Increased Plasma levels of CCBs

Metabolism: enzyme induction

- Cytochrome P450 isoenzymes mediate metabolic pathway of phase I oxidation.

- Enzyme induction interactions take 2 to 3 weeks to develop completely and are slow to resolve after stopping the enzyme inducer.

- Enzyme induction is a common mechanism of interaction and is also caused by the chlorinated hydrocarbon insecticides such as dicophane and lindane, and smoking tobacco.

- If one drug reduces the effects of another by enzyme induction, it may be possible to accommodate the interaction simply by raising the dose of the drug affected, but this requires good monitoring.

Metabolism: enzyme inducer

- The main drugs responsible for induction of the cytochrome P450 isoenzymes are

- Griseofulvin
- Phenytoin
- Rifampicin
- St. Johns wort
- Carbamazepine
- Phenobarbitone
- Cigerette Smoke

- Decreased concentration of warfarin, quinidine, cyclosporine, losartan, oral contraceptives and methadone

- GPRS Cell Phone

Metabolism: enzyme inhibition

•Enzyme inhibition is more common than enzyme induction.
•Enzyme inhibition results in to reduced metabolism of an affected drug and produces toxicity.
•Enzyme inhibition can occur within 2 to 3 days, resulting in the rapid development of toxicity.

•The metabolic pathway that is most commonly inhibited is phase I oxidation by cytochrome P450 isoenzymes.
•If the serum levels remain within the therapeutic range the interaction may not be clinically important.

Metabolism: enzyme inhibitor

- The main drugs responsible for inhibition of the cytochrome P450 isoenzymes

- Sulphonamides
- Antifungals (Itraconazole, Ketoconazole,Fluconazole)
- Macrolide Antibiotics (Clarithromycin,Azithromycin, Erythromycin)
- Ciprofloxacin
- Sertraline
- Cimetidine
- Omeprazole
- Metronidazole
- Antivirals (Ritonavir, Indinavir, Nelfinavir,Amprenavir)
- Antiarrhythmics (Amiodarone, Quinidine)
- Antidepressants (Fluoxetine, Paroxetine)
- Isoniazid
- Alcohol

- SICK FACES.COM

Metabolism: enzyme inhibitor

- Ketoconazole, Erythromycin, or Grapefruit juice

- Inhibition of CYP3A4
    - Blocks metabolism of Terfenadine
    - Increased plasma levels of Terfenadine
    - Fatal cardiac arrhythmias (Torsades de pointes)

Metabolism: enzyme inhibitor

- Gemfibrozil (and other fibrates)
    - CYP3A inhibition
    - Prevents metabolism of Statins (HMG-CoA reductase inhibitors)
    - Increased plasma levels
    - Rhabdomyolysis

Metabolism: enzyme inhibitor

- Ritonavir, indinavir, nelfinavir, amprenavir or saquinavir

- Inhibit CYP3A4
  - Blocks metabolism of Phosphodiesterase type-5 inhibitors (Sildenafil, Tadalafil, Vardenafil)
  - Increased serum levels PDE5 inhibitors

Metabolism: enzyme inhibitor

- Cimetidine
    - Inhibitor of multiple CYPs
    - Increased concentration of warfarin, theophylline and phenytoin
    - Toxicity

Metabolism: enzyme inhibitor

- **Amiodarone**
  - **Inhibitor of many CYPs and of P-glycoprotein**
  - **Decreased clearance (risk of toxicity) for warfarin, digoxin and quinidine**

Metabolism: Genetic factor

•Some of the population have a variant of the isoenzyme with different (usually poor) activity.

•a small proportion of the population have low CYP2D6 activity and are described as poor or slow metabolisers (about 5 to 10 in white Caucasians, 0 to 2 in Asians and black people).

•CYP2D6, CYP2C9 and CYP2C19 also show polymorphism, whereas CYP3A4 does not.

Metabolism:
Cytochrome P450 isoenzymes and predicting drug interactions

- If a new drug is shown to be an inducer, or an inhibitor, and/or a substrate of a given isoenzyme, we can predict likely drug interactions by using the list of enzyme inducer/inhibitor/substrate.

•For example, cyclosporin is metabolized by CYP3A4, and rifampicin is a known, potent inducer of this isoenzyme, whereas ketoconazole inhibits its activity.

  •Hence, rifampicin reduces the levels of ciclosporin and ketoconazole increases them.

PK-drug interaction: excretion

- Drug excretion interactions
    - Changes in urinary pH
    - Changes in active renal tubular excretion
    - Changes in renal blood flow
    - Biliary excretion and the entero-hepatic shunt
    - Enterohepatic recirculation.
    - Drug transporter proteins.

Excretion : Changes in urinary pH

•The pH of urine and the pKa of drugs determine the reabsorption of drugs.

•Only the non-ionised form is lipid-soluble and able to diffuse back through the lipid membranes of the renal tubular cells.

•The pH changes of urine such as alkaline urine for acidic drugs, acidic urine for basic drugs will increase the loss of the drug.

•Whereas alkaline urine for basic drugs, acidic urine for acidic drugs will increase their retention.

Excretion : Changes in urinary pH- weak base drug

- Weakly basic drugs (pKa 7.5 to 10.5)
- Exists as non-ionised lipid soluble molecules, at alkaline pH (high pH values)
    - Diffuse into the tubule cells
    - Reabsorption
    - Retention of drugs

Elimination: Changes in urinary pH- weak base drug

- Quinidine VS Antacids or Urinary Alkalinisers

- Quinidine
  - Excreted unchanged in urine
  - Antacids and urinary alkalinisers increase the pH of urine
  - Quinidine becomes non-ionised (Lipid soluble) in basic urine (High pH)
  - Diffuse into the tubule cells
  - Reabsorption
  - Reduced clearance of Quinidine

Elimination: Changes in urinary pH- weak acid drug

- Weak acids at Alkaline pH

- Weakly acid drugs (pKa 3 to 7.5)
  - Exists as ionised lipid-insoluble molecules, at alkaline pH (high pH values)
  - Unable to diffuse into the tubule cells
  - Remain in the urine
  - Removed from the body

Elimination: Changes in urinary pH- weak acid drug

- Aspirin and Antacids

- Aspirin
  - Antacids increase the pH of urine
  - Aspirin becomes ionised (Lipid insoluble) in basic urine (High pH)
  - Unable to diffuse into the tubule cells
  - Reabsorption is prevented
  - Increased urinary excretion
  - Helpful to treat overdose

Elimination: Changes in urinary pH- weak acid drug

- Methotrexate
  - Urinary alkalinisers increase the pH of urine
  - Methotrexate becomes ionised (Lipid insoluble) in basic urine (High pH)
  - Unable to diffuse into the tubule cells
  - Reabsorption is prevented
  - Increased urinary excretion
  - Helpful to treat Overdose

Elimination: active renal tubular excretion

- The competition between two or more drugs for same active transport systems in the renal tubules may affect the excretion.

- Probenecid
  - Inhibition of organic anion transporters (OATs)
    - Inhibition of renal secretion of anionic drugs (Cephalosporins, Dapsone, Methotrexate, Penicillins, Quinolones)
    - Increased serum levels
    - Toxicity of anionic drugs

Elimination: active renal tubular excretion

- Salicylates and some other NSAIDs
  - Inhibition of organic anion transporters (OATs)
    - Inhibition of renal secretion of Methotrexate
    - Increased serum levels
    - Toxicity of Methotrexate

Elimination: changes in renal blood flow

- The production of renal vasodilatory prostaglandins partially controls the flow of blood through the kidney.

- If the synthesis of these prostaglandins is inhibited, the renal excretion of some drugs may be reduced.

- NSAIDs
    - Inhibition of synthesis of the renal prostaglandins (PGE2)
    - Reduction of renal blood flow
    - Decreased renal excretion of the lithium
    - Lithium toxicity

imination: Biliary excretion and the entero-hepatic shunt

- A number of drugs are excreted in the bile, either unchanged or conjugated.

- Some of the conjugates
  - Metabolised to the parent compound by the gut flora
  - Reabsorbtion
  - Prolongation of stay of the drug within the body

- But if the gut flora are diminished by the presence of an antibacterial,
  - the drug is not recycled and is lost more quickly.

Elimination: Biliary excretion and the entero-hepatic shunt

- **Estrogen contraceptives + Penicillins**

- **The estrogen component of the contraceptive undergoes enterohepatic recirculation (i.e. it is repeatedly secreted in the bile as sulfate and glucuronide conjugates, which are hydrolyzed by the gut bacteria before reabsorption).**

- **Penicillin (Antibacterial)**
  - **Suppression of gut bacteria**
  - **Absence of hydrolysis of steroid conjugates**
  - **Poor reabsorption**
  - **Reduced concentrations of estrogen**
  - **Inadequate suppression of ovulation**

Elimination: Biliary excretion and the entero-hepatic shunt

- **Many drug transporter proteins (both from the ABC family and SLC family) are involved in the hepatic extraction and secretion of drugs into the bile.**

- **The bile salt export pump (ABCB11) is inhibited by the drugs like cyclosporin, glibenclamide,**

    - **Bosentan VS Glibenclamide or Ciclosporin**

        - **Inhibition of bile salt export pump (ABCB11)**
        - **Increased risk of cholestasis**

Pharmacodynamic drug interaction

- Pharmacodynamic interactions are those in which the effect of one drug is changed by the presence of another drug acting at the same biochemical or molecular site, on the same target organ, or on a different target.

- These interactions are classified as
    - Receptor mediated
    - Non receptor mediated

PD-DI: receptor mediated

- The therapeutic benefit of one or more drugs in the mix is diminished due to the interactions occurring at receptor level.

- Oxybutynin (Anticholinergic)
    - Blocks cholinergic receptors
    - Reduction of efficacy of Donepezil (Cholinesterase inhibitor)

PD-DI: non-receptor mediated

- Non-receptor mediated interactions produce additive or opposing therapeutic actions at a more systemic level.

- Sulfonylureas
    - Stimulate pancreatic insulin release
    - Hypoglycemia
    - Effective control of hyperglycemia is achieved by combining with biguanide drugs (metformin) which decreases hepatic glucose production

PD-DI: non-receptor mediated

- Aspirin VS Heparin

- Aspirin
    - Inhbits platelet aggregation
    - Increase the risk of bleeding when combined with Heparin (Anticoagulant)

PD-DI: non-receptor mediated

- **Furosemide VS Digoxin**

- **Furosemide**
    - **Secondary loss of potassium (Hypokalemia)**
    - **Increased Digoxin binding**
    - **Digoxin toxicity**

PD-DI: non-receptor mediated

- **Trimethoprim VS. ACEIs or ARBs**

- **Trimethoprim (Co-trimoxazole)**
  - **Structurally and pharmacologically similar to potassium-sparing diuretics**
    - **Induce hyperkalemia**
    - **Combining with ACEIs or ARBs**
    - **Increased risk of Hyperkalemia**

Classification of Pharmacodynamic drug interaction

- **Additive (Summation)**
  - combined effect of two chemicals is equal to the sum of individual effects of them
- **Synergistic**
  - the effect of two chemicals taken together is greater than the sum of their separate effect
- **Potentiation**
  - the creation of a toxic effect from one drug due to the presence of another drug
- **Antagonism**
  - the effect of two chemicals is actually less than the sum of the effect of the two drugs taken independently
    - Functional
    - Chemical
    - Dispositional
    - Receptor

Additive Interaction

- **Additive or Summation interaction occurs, when two or more drugs that produce similar physiologic effects are administered.**

- **Additive interaction means the combined effect of two chemicals is equal to the sum of individual effects of them.**
- **i.e. 1 + 1= 2**

- **The additive effects can produce excessive response or toxicity.**

- **Examples**
    - **Aspirin + Ibuprofen**

Additive interaction

- **Additive effects can occur with both the main effects of the drugs as well as their adverse effects.**

- **Ex Antimuscarinic antiparkinson drugs (main effect) + Butyrophenones (adverse effect)**
  - **Serious antimuscarinic toxicity**

Synergistic Interaction

- **Synergistic interaction means that the effect of two chemicals taken together is greater than the sum of their separate effect at the same doses. i.e Synergistic effect 1 + 1 > 2**

- **Synergism, in which the action of one drug enhances the action of another.**

- **Examples**
- **Aminoglycoside + Penicillin**
    - **Increased antibacterial activity (Beneficial effect)**

- **Barbiturates + Alcohol**
    - **Increased CNS depression (Harmful effect)**

Potentiation interaction

- Potentiation describes the creation of a toxic effect from one drug due to the presence of another drug. i.e Potentiation effect 1+ 0 = toxic

- Examples
  - Fluoroquinolones + Macrolides
    - Excessive QT prolongation
    - Torsades de pointes

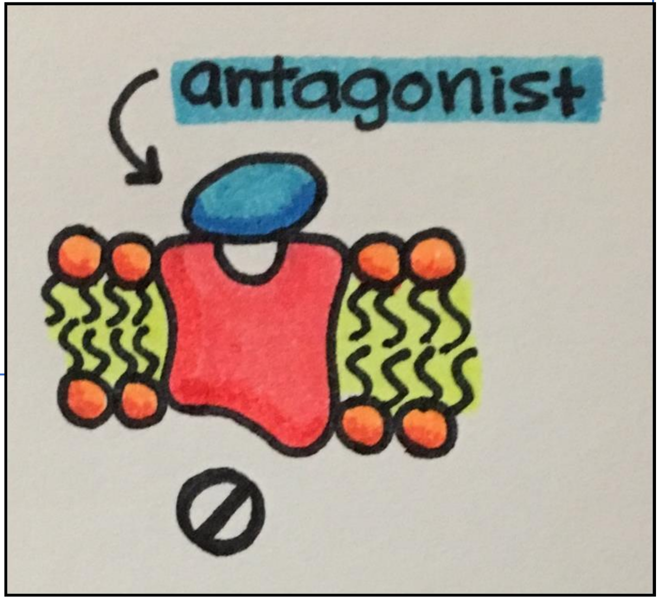
- ACE inhibitors + Potassium-sparing diuretics(Amiloride)
  - Increased potassium retention
  - Life-threatening hyperkalemia

Antagonistic interaction

- **Antagonistic interaction means that the effect of two chemicals is actually less than the sum of the effect of the two drugs taken independently of each other. i.e Antagonism 1 +1= 0**

- **Antagonism is also defined as the interference of one drug with the action of another.**
- **Antagonism reduces the pharmacological effect of one drug (agonist) by a second drug**
- **Antagonism forms the basis for antidotes of poisonings.**

- **Types of Antagonism**
    - **Functional or Physiological**
    - **Chemical or Inactivation**
    - **Dispositional**
    - **Receptor**



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Antagonism-Functional antagonist

- **Functional or physiological antagonism** occurs when two chemicals **produce opposite effects on the same physiological function**.
- Physiological antagonism describes the behavior of a substance that counteracting effects of another substance **without binding to the same receptor**.
- This is the basis for most supportive care provided to patients treated for drug overdose poisoning.

- **Epinephrine (Adrenaline)**
  - Alpha 1-adrenergic receptor activation
  - Vasoconstriction
  - Raised arterial pressure in contrast to histamine, which lowers arterial pressure.
  - Epinephrine and other such substances are physiological antagonists to histamine and they do not bind to and block the histamine receptor.

Antagonism-Chemical antagonist

•Chemical antagonism, or inactivation, is a **reaction between two chemicals to neutralize their effects.**

•A drug counters the effect of another by simple chemical reaction or neutralization, in chemical antagonism.

•Examples

•1. Protamine sulphate (Basic) VS. Heparin(Acidic)

- •Protamine sulphate is antidote for Heparin overdosage

•2. Dimercaprol (Chelating agent) VS. Mercury

- •Dimercaprol is useful in Mercury poisoning.

Antagonism-Dispositional antagonist

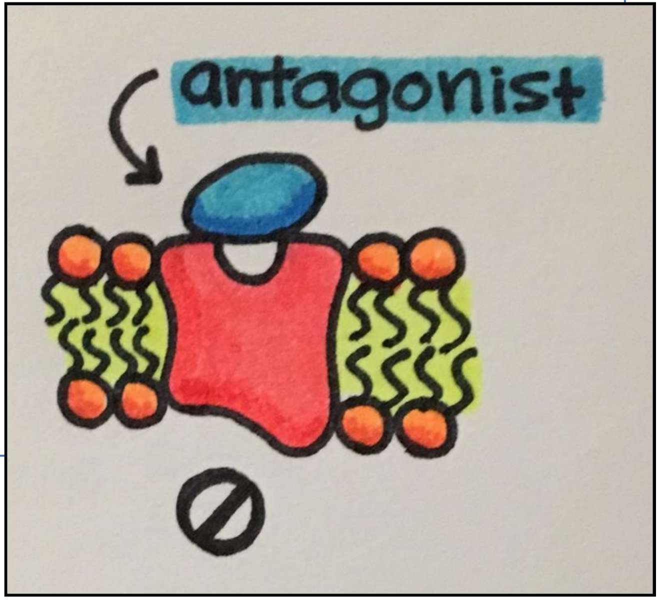
- Dispositional antagonism is the alteration of the disposition of a substance (its absorption, biotransformation, distribution, or excretion).
    - less of the agent reaches the target organ or its persistence in the target organ is reduced.

- Ex
    - Cholestyramine
        - Binds to bile acids
        - Prevention of reabsorption of bile acids
        - Production of Insoluble complex
        - Excreted in the feces

Antagonism-Receptor antagonist

- **Receptor antagonism involves the blockade of the effect of a drug (Agonist) with another drug (Antagonist) that competes at the receptor site.**

- **Types of Receptor antagonists**
    - **Competitive antagonists**
        - **Reversible competitive antagonists**
        - **Irreversible competitive antagonists**
    - **Non competitive antagonists**
    - **Uncompetitive antagonists**
    - **Partial agonists**
    - **Inverse agonists**



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Antagonism-Receptor antagonist: competitive

- The antagonist will compete with available agonist for receptor binding sites on the same receptor.

- Competitive antagonists
    - Bind to receptors at the same binding site of endogenous ligand or agonist
    - Displace the agonist from the binding sites
    - Lower frequency of receptor activation

- Example
    - Naloxone is a competitive antagonists at all opioid receptors.

Antagonism-Receptor antagonist: competitive

- Reversible competitive antagonism
  - This type of antagonist can be reversed with an increase in concentration of agonist.
    - Ex Atropine blocks Mucarinic Ach receptors reversibly.

- Irreversible competitive antagonism
  - This type of antagonist cannot be reversed without a complete regeneration of the system.
    - Ex Phenoxybenzamine blocks adrenergic alpha-receptors and is used to treat pheochromocytoma.

Antagonism-Receptor antagonist: non-competitive

- An antagonist that binds to the allosteric site of a receptor.
    - Non-competitive antagonists reduce the magnitude of the maximum response.

- Non-competitive antagonists
    - Do not compete with agonists for binding at the active site
    - Bind to a distinctly separate binding site from the agonist
    - Prevent conformational changes in the receptor
    - Blockade of receptor

- Example Ketamine (Non competitive antagonist) binds in the NMDA receptor channel pore but the Glutamate (Agonist) binds to the extracellular surface of the receptor.

Antagonism-Receptor antagonist: uncompetitive

- Uncompetitive antagonists bind to a separate allosteric binding site.

1. Agonists
2. Activation of receptors
3. Binding of Uncompetitive antagonists
4. Blockade of receptors
    - The blockade is better in higher concentrations of agonist than lower concentrations of agonist.

- Example Memantine uncompetitively blocks NMDA receptor and is used in the treatment of Alzheimer's disease.

Antagonism-Receptor antagonist: partial-agonist

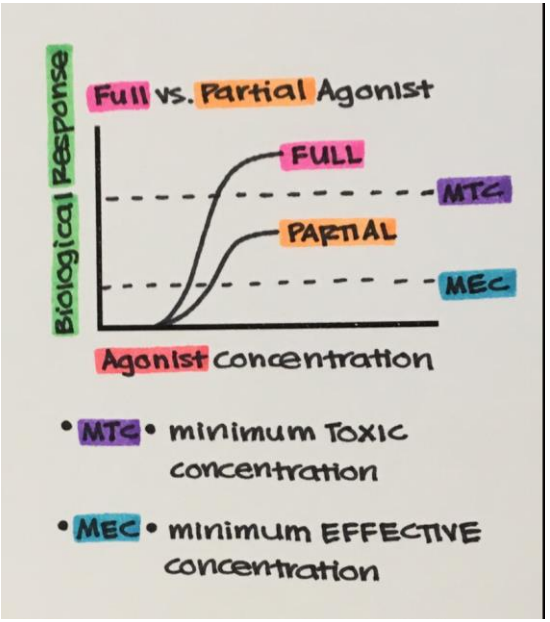
•Partial agonists can act as a competitive antagonist in the presence of a full agonist.

•Partial agonists
- •Compete with the full agonist for receptor occupancy
- •Net decrease in the receptor activation
- •Clinically, their usefulness is derived from their ability to enhance deficient systems while simultaneously blocking excessive activity.

•Exposing a receptor to a high level of a partial agonist will ensure that it has a constant, weak level of activity.

•Partial agonism prevents the adaptive regulatory mechanisms that frequently develop after repeated exposure to potent full agonists or antagonists.

•Example Buprenorphine, a partial agonist of the μ-opioid receptor, with weak morphine-like activity and is used clinically as an analgesic in pain management and as an alternative to methadone in the treatment of opioid dependence.



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Antagonism-Receptor antagonist: inverse agonist

- **Inverse agonists**
    - **Bind to Constitutive active receptors**
    - **Block the effects of binding agonists and also inhibit the basal activity of the receptor**

เอกสารอ้างอิง

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